

Qatar-1b

R-0017

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_260601 · created 2026-07-08T08:03:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461192.8272 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.057
Transit depth (Rp/Rs)^2	3.55 +/- 2.13 [%]
Orbital Inclination (inc)	82.91 +/- 2.87
Airmass coefficient 1 (a1)	3.29 +/- 2.15
Airmass coefficient 2 (a2)	-0.4 +/- 0.83
Scatter in the residuals of the lightcurve fit is	54.22 %
Best Comparison Star	#2 - [431, 458]
Optimal Aperture	2.41
Optimal Annulus	3.5
Transit Duration (day)	0.059 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.188 $\pm$ 0.057 vs 0.1463 (deviation 0.56 σ)
Duration fit vs published	0.059 d vs 0.0692 d (deviation 0.33 σ)
Mid-time O-C	-1.9 min
Depth SNR	2.5
Residual scatter	54.22 %

Light curve

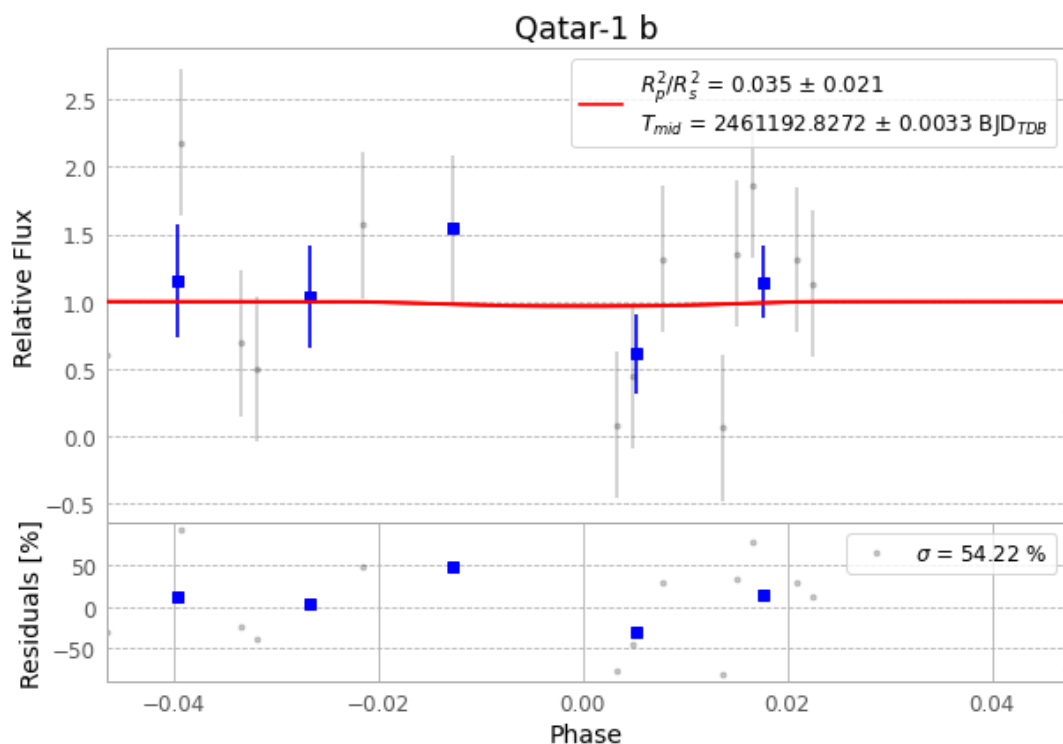


Figure 1 — FinalLightCurve_Qatar-1 b_2026-06-01.png (SHA-256 f3a3df938dd082e3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9a7720c323de8099...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

74 FITS frames (49 MB), Qatar-1260601060316.FITS → Qatar-1260601094216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260601.log (965cf276b59e...), FinalLightCurve_Qatar-1 b_2026-06-01.png (f3a3df938dd0...), FinalLightCurve_Qatar-1 b_2026-06-01.csv (1222e579606a...)

Compute resources

Wall time 149.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.